ARM® Cortex®-M23

32-bit Microcontroller

NuMicro® Family
M2U51 Series BSP
Revision History

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Revision 3.01.000 (Released 2026-10-14)

1. Fix Doxygen warnings, comments, and typos
2. Merge "Update fmc.h to fix misra issue"
3. Update fmc.h to fix misra issue
4. Modify some drivers to fix the MISRA violation.
5. Modify some drivers to fix the MISRA violation.
6. Fix VSCode FMC_APWPROT sample code build error
7. Add XOM & APWPROT sample code
8. Merge "Fix FMC_ExeInSRAM scatter file comments to avoid ARMCLANG warning"
9. Fix FMC_ExeInSRAM scatter file comments to avoid ARMCLANG warning
10. Merge "Modify gpio.c to fix the GPIO related MISRA violation"
11. Merge "Update linker loading file to fix build error"
12. Update linker loading file to fix build error
13. Modify gpio.c to fix the GPIO related MISRA violation
14. Merge "[SBOM] Complete FreeRTOS and firmware review evidence"
15. Adjust FMC_ExeInSRAM memory layout to resolve insufficient space
16. [SBOM] Complete FreeRTOS and firmware review evidence
17. [SBOM] Synchronize V3.00.000-14 release artifacts
18. [SBOM] Refresh current SBOM for source review
19. [SBOM] Synchronize V3.00.000-12 release artifacts
20. [SBOM] Update current SBOM for FreeRTOS
21. [SBOM] Add FreeRTOS component evidence
22. Merge "Add FreeRTOS sample code"
23. Add FreeRTOS sample code
24. Merge "M2U51: Move GCC HardFault handler from retarget to startup for consistent toolchain handling"
25. [SBOM] Synchronize V3.00.000-5 release artifacts
26. M2U51: Move GCC HardFault handler from retarget to startup for consistent toolchain handling
27. [SBOM] Update current Product and Test Sample SBOMs
28. [SBOM] Add current Product and Test Sample SBOMs
29. [SBOM] Identify FMC IAP firmware licenses
30. [SBOM]Add canonical CMSIS component hashes
31. Refine Product scope and semihost disposition
32. Merge "Modify timer_pwm.h and timer_pwm.c to fix MISRA violation."
33. Modify timer_pwm.h and timer_pwm.c to fix MISRA violation.
34. Merge "Fixed typos and remove unused file."
35. Merge "Modify bpwm/pwm/spi/timer/timer_pwm/lcd dirvers for MISRA checking"
36. Fixed typos and remove unused file.
37. Modify bpwm/pwm/spi/timer/timer_pwm/lcd dirvers for MISRA checking
38. Correct RTC reference definition for frequency compensation
39. Update startup, system, and retarget files
40. Add SBOM files
41. Fix CRC/CRYPTO/FMC/RTC/WDT MISRA C violations
42. Fix MISRA-C violations for I2C, PDMA, UART and USCI driver.
43. Fix FMC_MultiBoot Build fail
44. Remove NPD2 feature.
45. update README.md
46. Add an LCD_RTC_Alarm_Wakeup sample to demonstrate how to use the RTC to wake the system from Power-down mode and update the current time on the LCD

47. MOdify several LCD sample codes to meet with the new 'NuMaker-TNLCDSub_M254K_4.8V_V1.1' LCD daughter board
48. Modify PDMA EnableInt function to support two more mask at the same time
49. Modify TIMER_ResetCounter() in timer.c
50. Fix GITHUB ACTION VSCode Build Error
51. Update Revision History & Driver Reference Guide
52. Fix FMC_IAP & MultiBoot VSCode Build Error

Revision 3.00.001 (Released 2025-11-05)

53. Fix ISP sample code VScode GCC build fail
54. Add VSCode Quick Start Guide.pdf
55. Remove VSCode project for sample code SYS_SPDMode_WakeupVTOR.
56. Define HardFault_Handler() for VSCode GCC project.
57. Remove VSCode project for sample code Semihost.
58. Merge "Fix bug about floating calculation."
59. Fix bug about floating calculation.
60. Update Template VSCode project
61. Merge "Add --specs=nosys.specs to VSCode projects"
62. Add --specs=nosys.specs to VSCode projects
63. Update GPIO driver for EINT"
64. Add -Wl,--no-warn-rwx-segments to VSCode projects
65. Fix VSCode GNUC build error
66. Fix VSCode GCC build error
67. Fix: Prevent pointer addition on NULL and array out-of-bounds access
68. Modify I2C/UI2C Sample codes for VSCode Supporting
69. Remove linker file setting from VSCode Project files
70. Correct comments
71. Update the VSCode project files of FMC_MultiBoot
72. Update IRQ setting & code for WWDT
73. Update Crypto IRQ handler definition name
74. Fix build warning & error to support GCC 14.3.1
75. Update launch.json, tasks.json, and vcpkg-configuration.json to use cortex-debug
76. Update system setting as following:
    - Set BGWKM to 1
    - Delay 100us when switch NRBGLPEL mode
77. Fixed ISP_I2C/ISP_SPI
78. Modify LCD sample codes to fix the bug in pin MFP settings.
79. Modify PWM driver to remove M4xx related description
80. Fix the building error for sample code ADC_PDMA_BPWM_Trigger.

Revision 3.00.000 (Released 2025-05-16)

81. Primary release version.

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